Glaze recipes, step by step: "sand satin"

Version 2026-11-07 · the 🧪 Glaze recipes page (card G) with the **Auto** engine, driven in a browser through every kind of input, from a colour alone to words + photo + formula. Every screenshot is the real page; every number is the page's own result (a test copy of the app with its own logs).

▭ red frame = where to look (the inside stays see-through) · orange = from **test_similarity.zip** (the team's colour-match model and its rules) · purple = from the **text branch** · black = the Glaze Search project

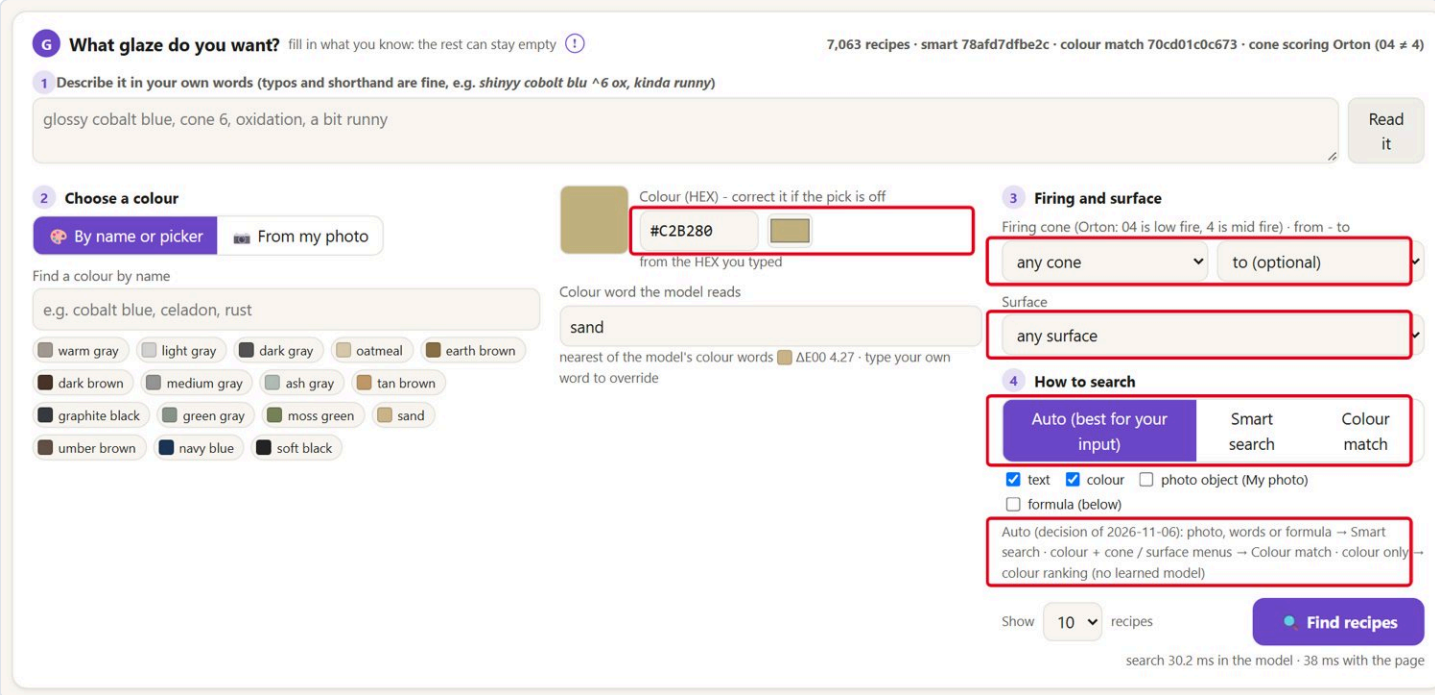
The three engines, and how Auto picks one

Auto uses	when you give	score of a recipe	models and data
Colour ranking	only a colour	0 × learned + **0.80 × colour** + 0.10 × surface + 0.10 × cone	the colour score exp(−ΔE00/20), surface and cone rules and the 6,718-recipe catalog; the learned part is switched off (no model)
Colour match	colour + cone / surface menus, no words	0.40 × learned + 0.40 × colour + 0.10 × surface + 0.10 cone	all-MiniLM-L6-v2 (frozen) reads the template `color: …, Cone …, …, unknown, unknown` → text head 384→256; recipes: 26 oxides → √ → PCA-16 → chemistry head 16→256; learned = (cos + 1)/2 in their 256-d latent space
Smart search	typed words, a photo object or a formula	its own weights (words 0.35 · photo 0.35 · formula 0.25 · colour 0.30 · cone 0.10 · surface 0.10 · transparency 0.05 · atmosphere 0.05), divided by the total of the parts you gave	MiniLM + DINOv2 (frozen) + oxides → trained heads in one 256-d space, trained with the team's method (masked InfoNCE, composition groups); recipe text = colour word + tags cleaned by the text branch

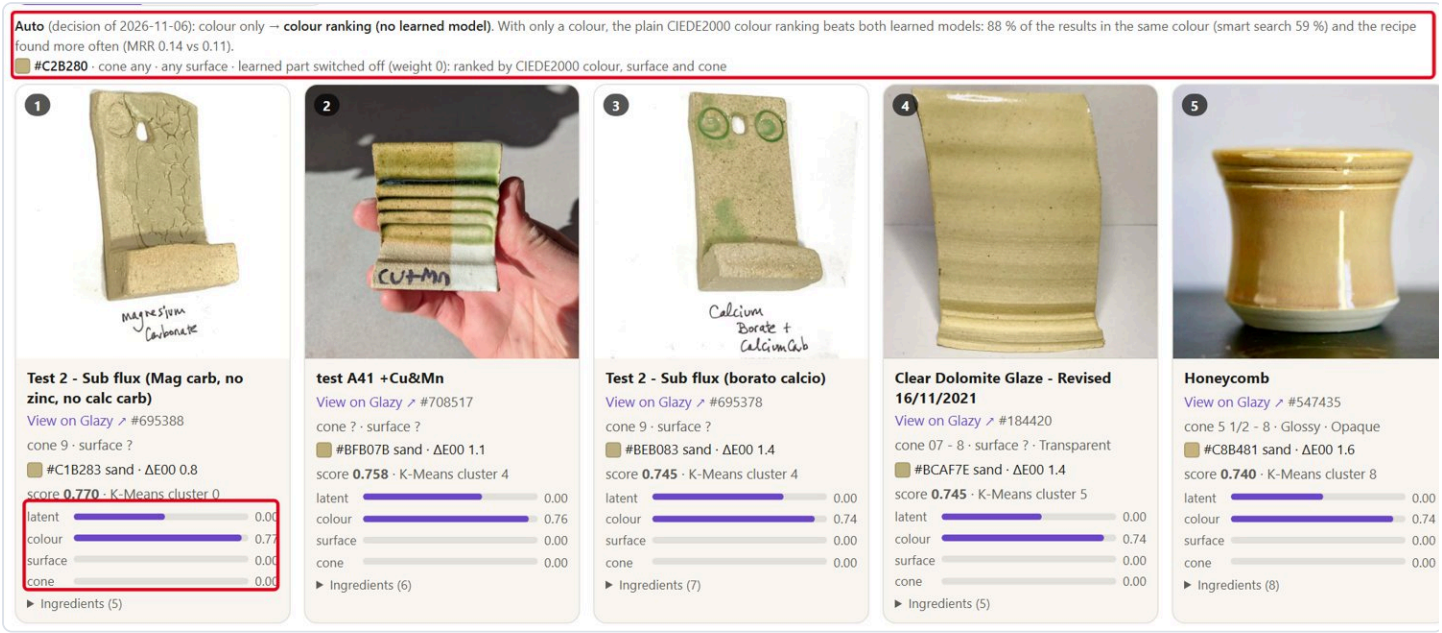
colour value = exp(−ΔE00 / 20) · learned / words / photo / formula value = (cosine + 1) / 2 · surface, cone = 1 when they match, 0 when not or unknown · score = Σ weight × value (0 – 1)

Case 1 · Only a colour → colour ranking

1. Leave the words empty; type the HEX `#C2B280` (sand) in 2 · Choose a colour.
2. Leave cone and surface on "any"; 4 · How to search stays on **Auto**. Press **Find recipes**.



Card G: the colour (framed), the empty menus, Auto and its rules (framed).



Auto chose colour ranking: the header says the learned part is switched off; #1's bars (framed): only colour counts.

The calculation of #1

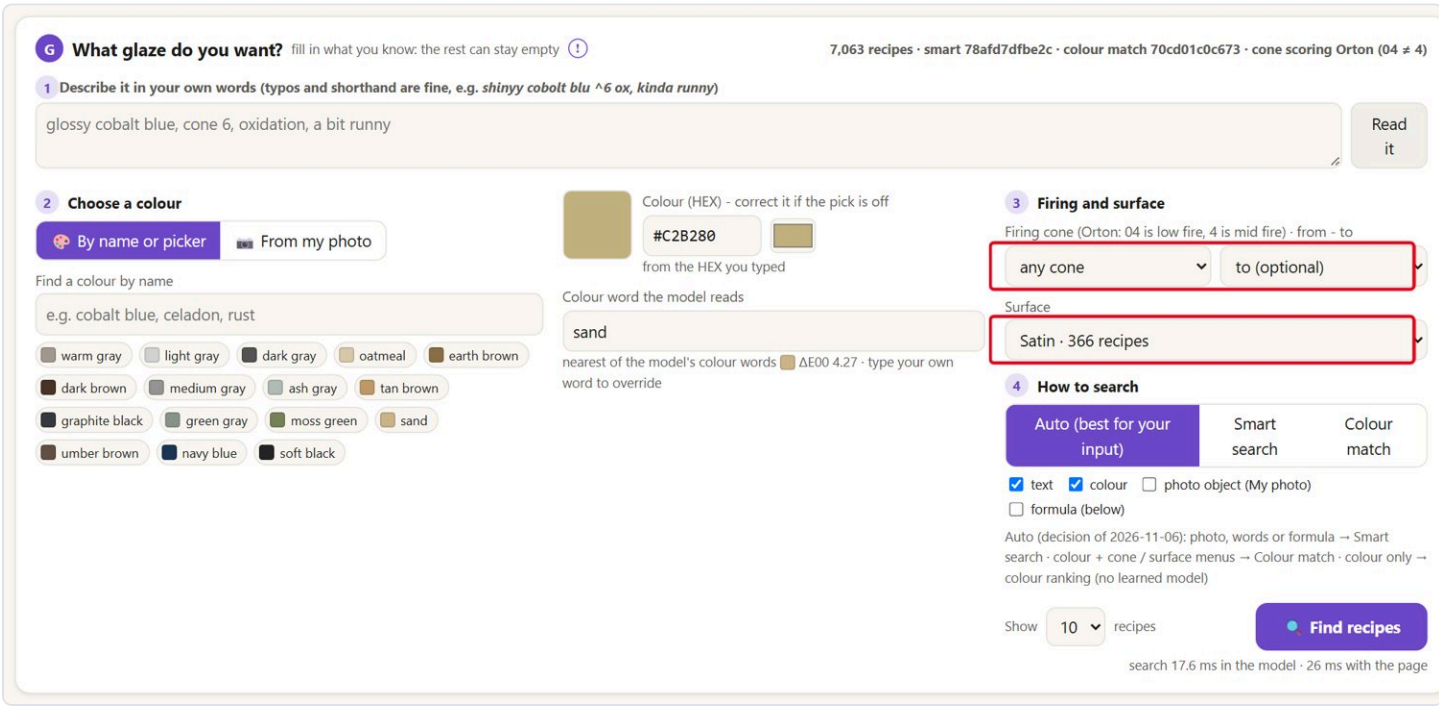
part	weight	value	weight × value	what the value is
learned match	0.000	0.523	**0.000**	(cos + 1) / 2 of the template sentence and the recipe's chemistry in the 256-d latent space
colour	0.800	0.963	**0.770**	exp(−ΔE00 / 20), ΔE00 = 0.76
surface	0.100	no data	**0.000**	no data on the recipe: counts 0
cone	0.100	no data	**0.000**	no data on the recipe: counts 0
score of #1 Test 2 - Sub flux (Mag carb, no zinc, no calc carb)			**0.770**	

- **Test 2 - Sub flux (Mag carb, no zinc, no calc carb)** · 0.770 · sand #C1B283 · cone 9 · surface ?
- **test A41 +Cu&Mn** · 0.758 · sand #BFB07B · cone ? · surface ?
- **Test 2 - Sub flux (borato calcio)** · 0.745 · sand #BEB083 · cone 9 · surface ?

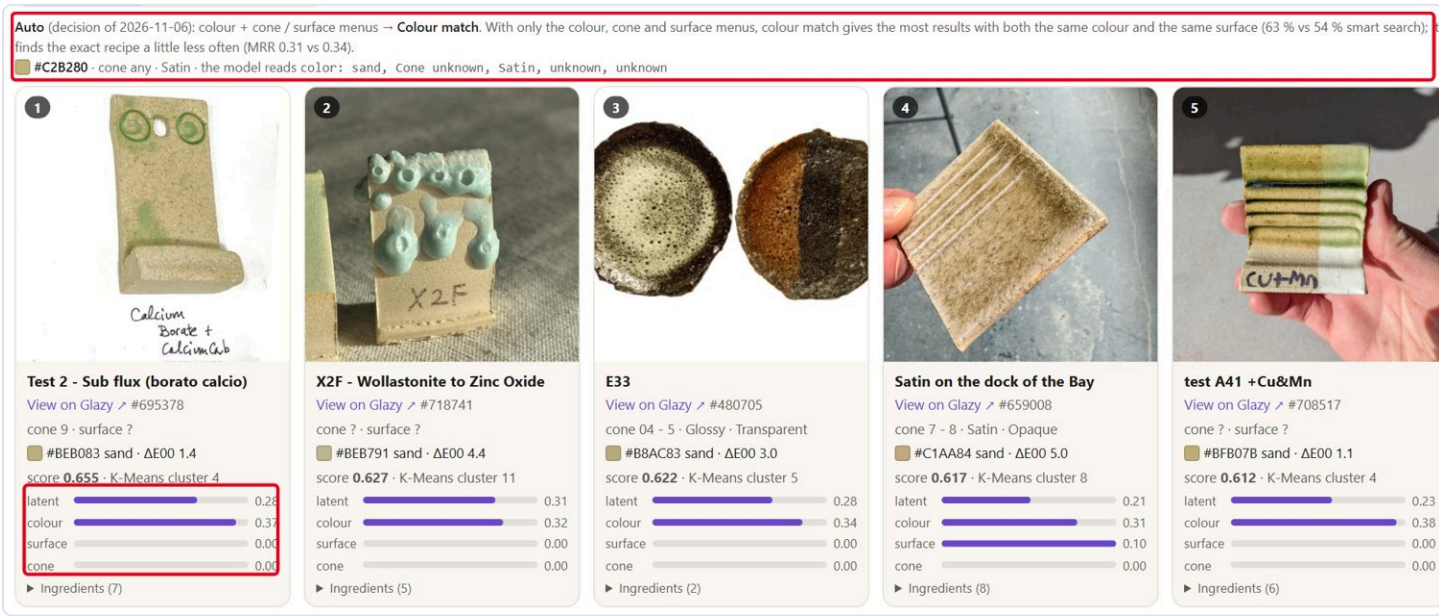
What it shows: the exact colour (ΔE00 under 2), but the cones (9) and surfaces (unknown) are whatever they happen to be: nothing asked for them.

Case 2 · Colour + the menus, no words → Colour match (the team's model, 4 / 4 / 1 / 1)

1. **2a:** keep the sand colour and choose surface **Satin** in 3 · Firing and surface. Find recipes.
2. **2b:** also choose cone **6**. Find recipes.



2a: surface Satin, cone still 'any' (framed).

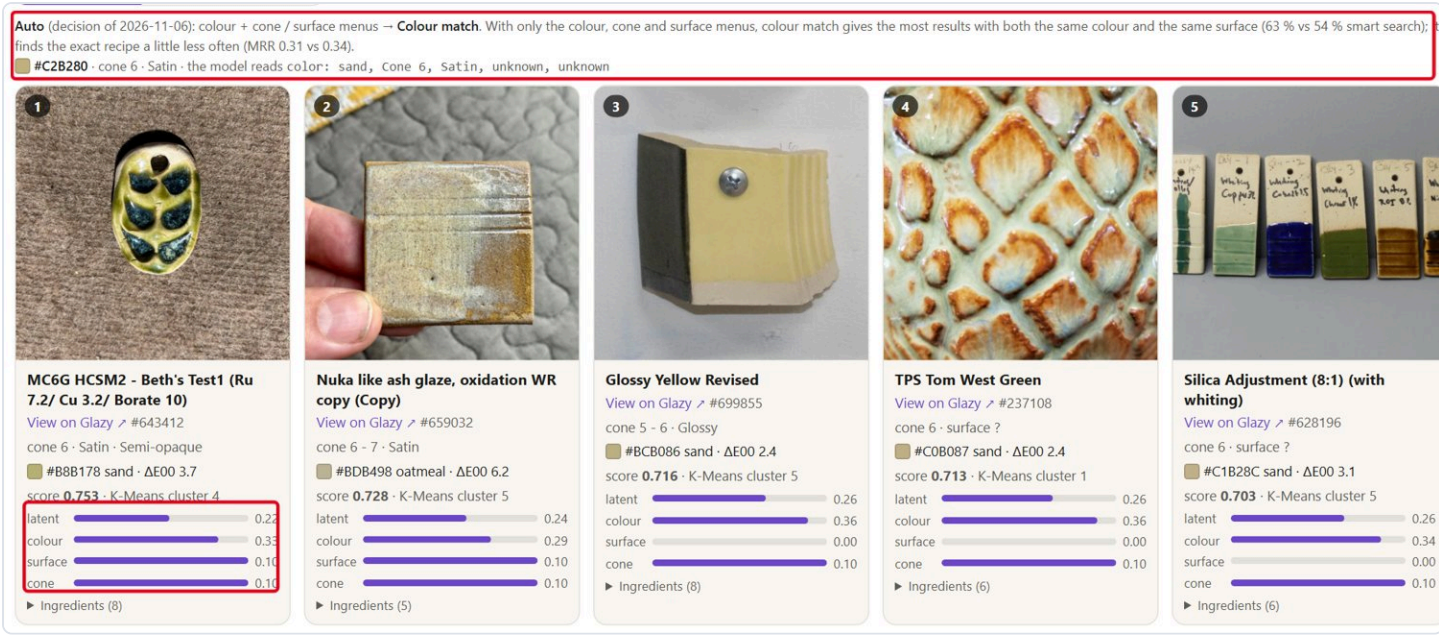


2a: Auto chose Colour match; it reads `color: sand, Cone unknown, Satin, unknown, unknown` (framed).

2a · the calculation of #1

part	weight	value	weight × value	what the value is
learned match	0.400	0.706	**0.282**	(cos + 1) / 2 of the template sentence and the recipe's chemistry in the 256-d latent space
colour	0.400	0.931	**0.372**	exp(−ΔE00 / 20), ΔE00 = 1.43
surface	0.100	no data	**0.000**	no data on the recipe: counts 0
cone	0.100	no data	**0.000**	no data on the recipe: counts 0
score of #1 Test 2 - Sub flux (borato calcio)			**0.655**	

#1 has no surface and no cone on record, so both count 0; it still wins on colour (ΔE00 1.4) and the learned match. That is why 2a mixes in cone-9 and unknown-surface recipes.



2b: cone 6 added: the template now says `Cone 6`, and #1's surface and cone bars are full (framed).

2b · the calculation of #1

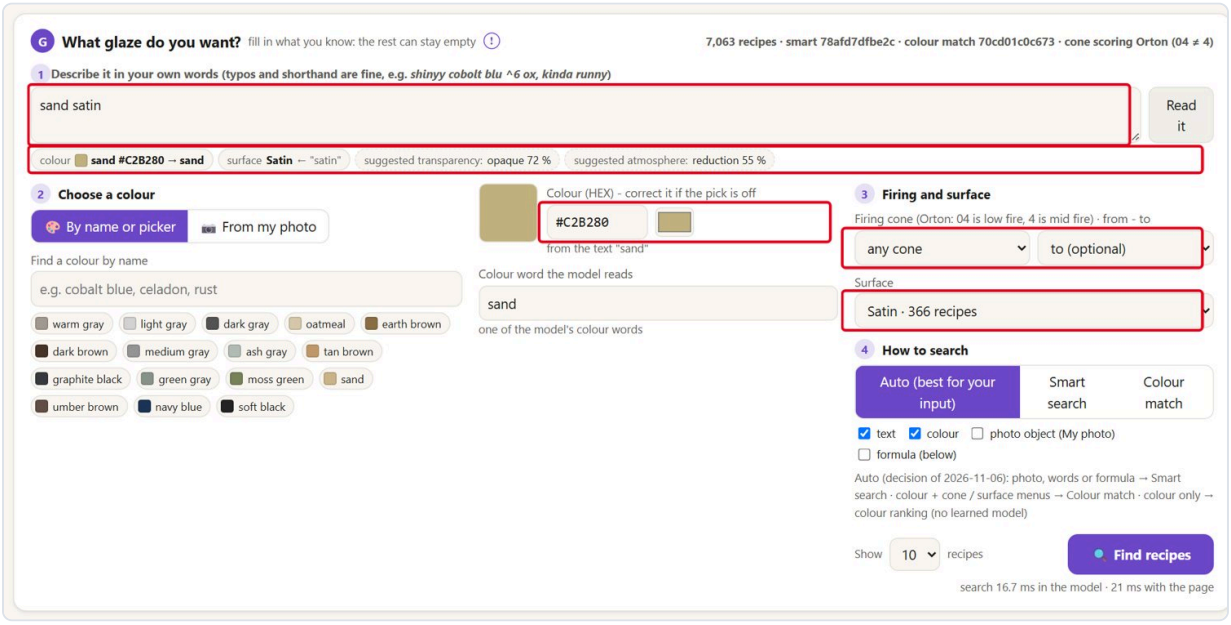
part	weight	value	weight × value	what the value is
learned match	0.400	0.553	**0.221**	(cos + 1) / 2 of the template sentence and the recipe's chemistry in the 256-d latent space
colour	0.400	0.830	**0.332**	exp(−ΔE00 / 20), ΔE00 = 3.71
surface	0.100	1.000	**0.100**	1 = matches, 0 = does not
cone	0.100	1.000	**0.100**	1 = matches, 0 = does not
score of #1 MC6G HCSM2 - Beth's Test1 (Ru 7.2/ Cu 3.2/ Borate 10)			**0.753**	

- **MC6G HCSM2 - Beth's Test1 (Ru 7.2/ Cu 3.2/ Borate 10)** · 0.753 · sand #B8B178 · cone 6 · Satin
- **Nuka like ash glaze, oxidation WR copy (Copy)** · 0.728 · oatmeal #BDB498 · cone 6 - 7 · Satin
- **Glossy Yellow Revised** · 0.716 · sand #BCB086 · cone 5 - 6 · Glossy

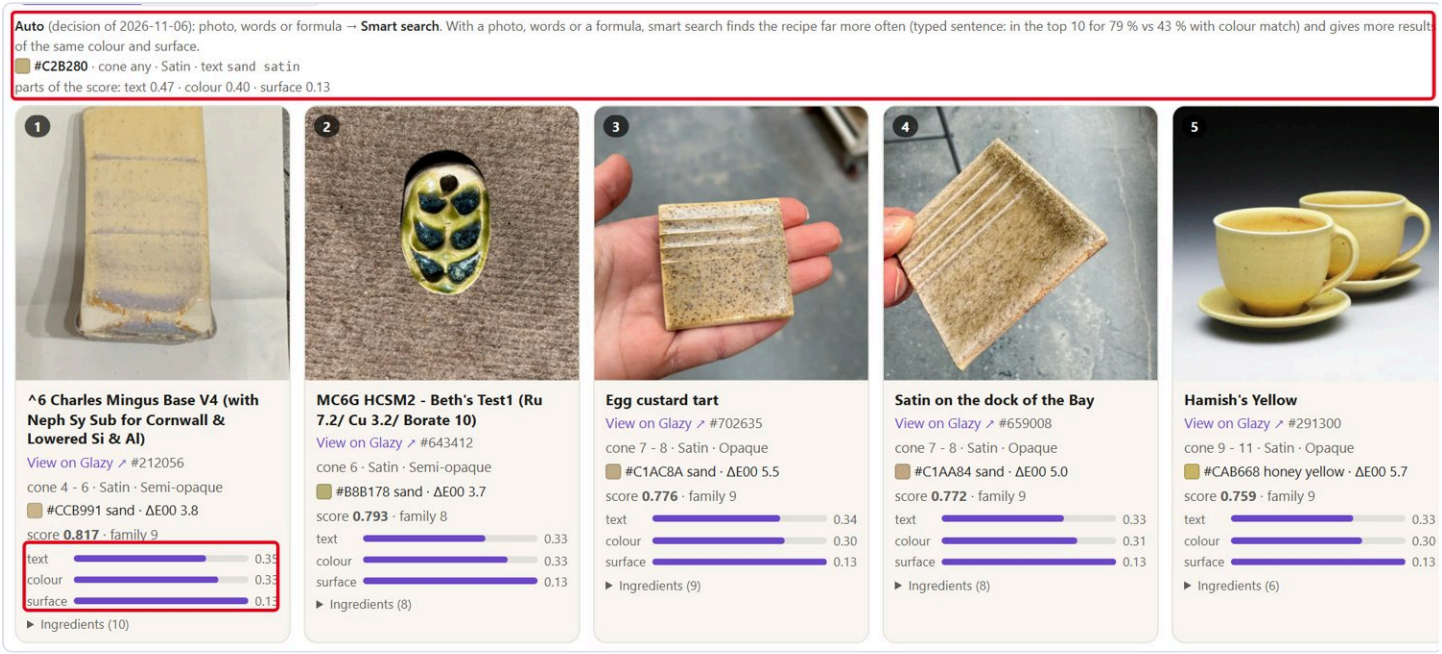
> **Always set the cone.** It fills the template (the learned part reads it) and the cone part of the score: in 2b all three results are cone 6 and the right colour family.

Case 3a · Typed words → Smart search

1. Set the menus back to "any" and type `sand satin` in 1 · Describe it. Press Find recipes: the page reads the words first.



Reading the words (framed): 'sand' sets the colour #C2B280, 'satin' sets the surface menu; Auto then picks Smart search.



Smart search: the parts of the score in the header, #1's bars (framed).

The calculation of #1

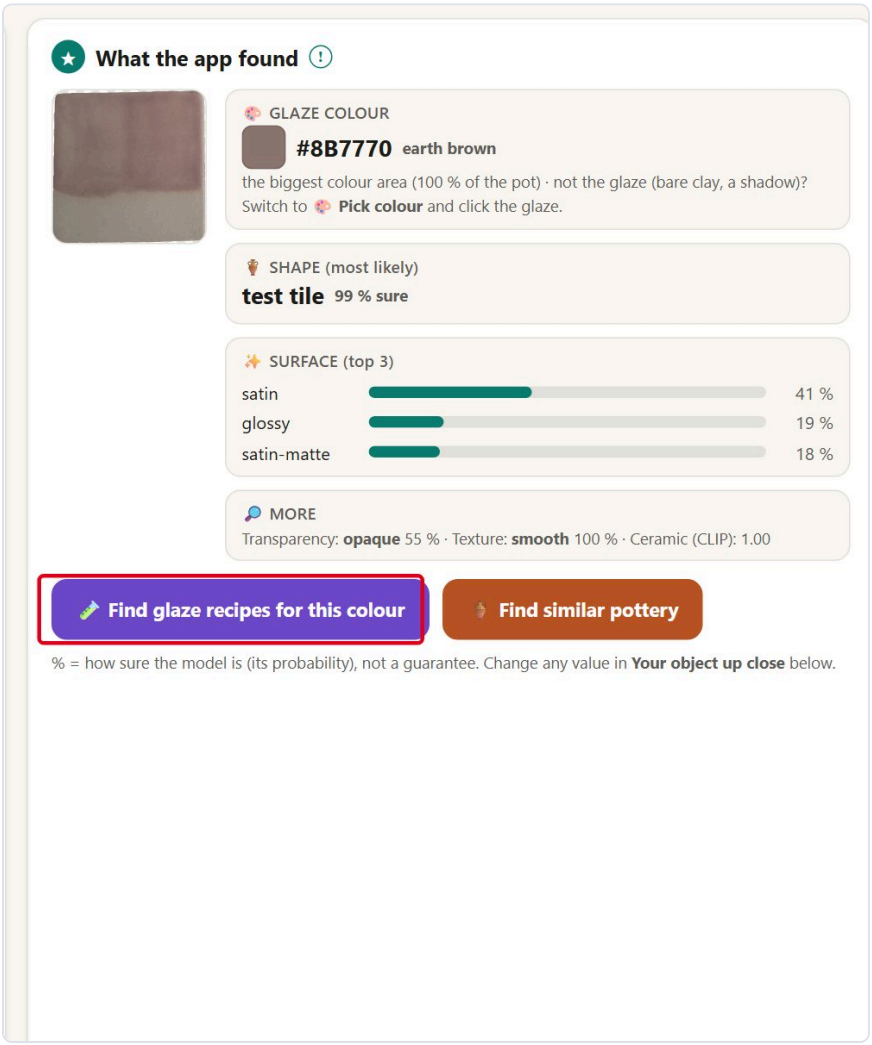
part	weight	value	weight × value	what the value is
words	0.467	0.757	**0.353**	(cos + 1) / 2 of your words and the recipe in the joint space
colour	0.400	0.825	**0.330**	exp(−ΔE00 / 20), ΔE00 = 3.84
surface	0.133	1.000	**0.133**	1 = matches, 0 = does not
score of #1 ^6 Charles Mingus Base V4 (with Neph Sy Sub for Cornwall & Lowered Si & Al)			**0.817**	

- **^6 Charles Mingus Base V4 (with Neph Sy Sub for Cornwall & Lowered Si & Al)** · 0.817 · sand #CCB991 · cone 4 - 6 · Satin
- **MC6G HCSM2 - Beth's Test1 (Ru 7.2/ Cu 3.2/ Borate 10)** · 0.793 · sand #B8B178 · cone 6 · Satin
- **Egg custard tart** · 0.776 · sand #C1AC8A · cone 7 - 8 · Satin

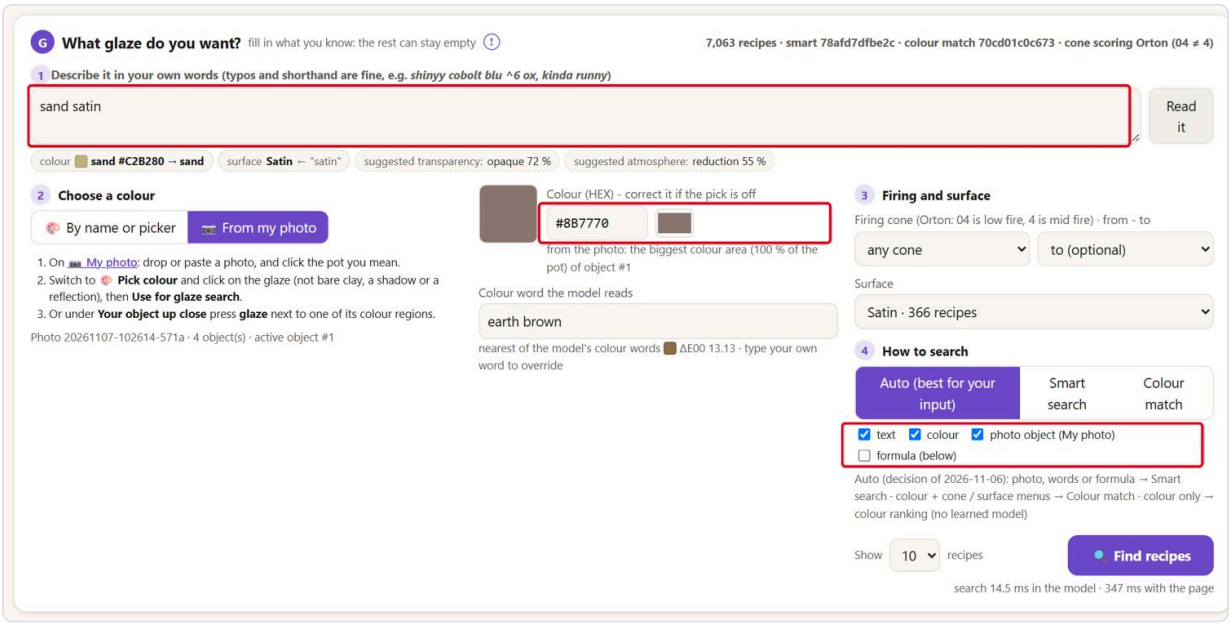
What it shows: without a photo, typing the words gives the best list: all three are sand-coloured and Satin.

Case 3b · + a photo → Smart search with the photo

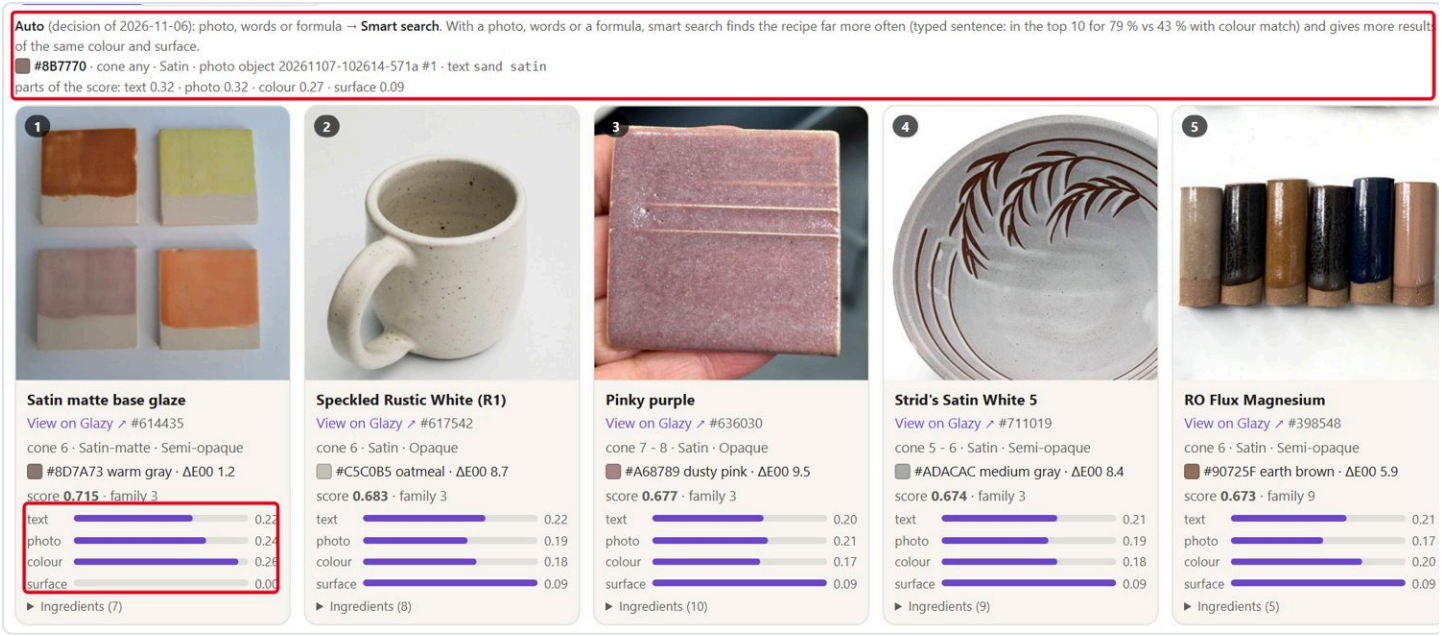
1. 📷 My photo: upload the Manual's test-tile photo (Glazy #614435). The app finds the tiles and takes the main colour of the selected one.
2. Press 💉 **Find glaze recipes for this colour**: the recipes page searches with the photo object, its colour and the words already typed.



What the app found, and the button that sends the photo to the recipe search (framed).



Card G now ticks 'photo object'; the colour is the photo's, #8B7770 (framed).



Smart search with words, photo, colour and surface (header framed).

The calculation of #1

part	weight	value	weight × value	what the value is
words	0.318	0.676	**0.215**	(cos + 1) / 2 of your words and the recipe in the joint space
photo	0.318	0.762	**0.243**	(cos + 1) / 2 of the cut-out photo and the recipe in the joint space
colour	0.273	0.944	**0.257**	exp(−ΔE00 / 20), ΔE00 = 1.16
surface	0.091	0.000	**0.000**	1 = matches, 0 = does not
score of #1 Satin matte base glaze			**0.715**	

- **Satin matte base glaze** · 0.715 · warm gray #8D7A73 · cone 6 · Satin-matte
- **Speckled Rustic White (R1)** · 0.683 · oatmeal #C5C0B5 · cone 6 · Satin
- **Pinky purple** · 0.677 · dusty pink #A68789 · cone 7 - 8 · Satin

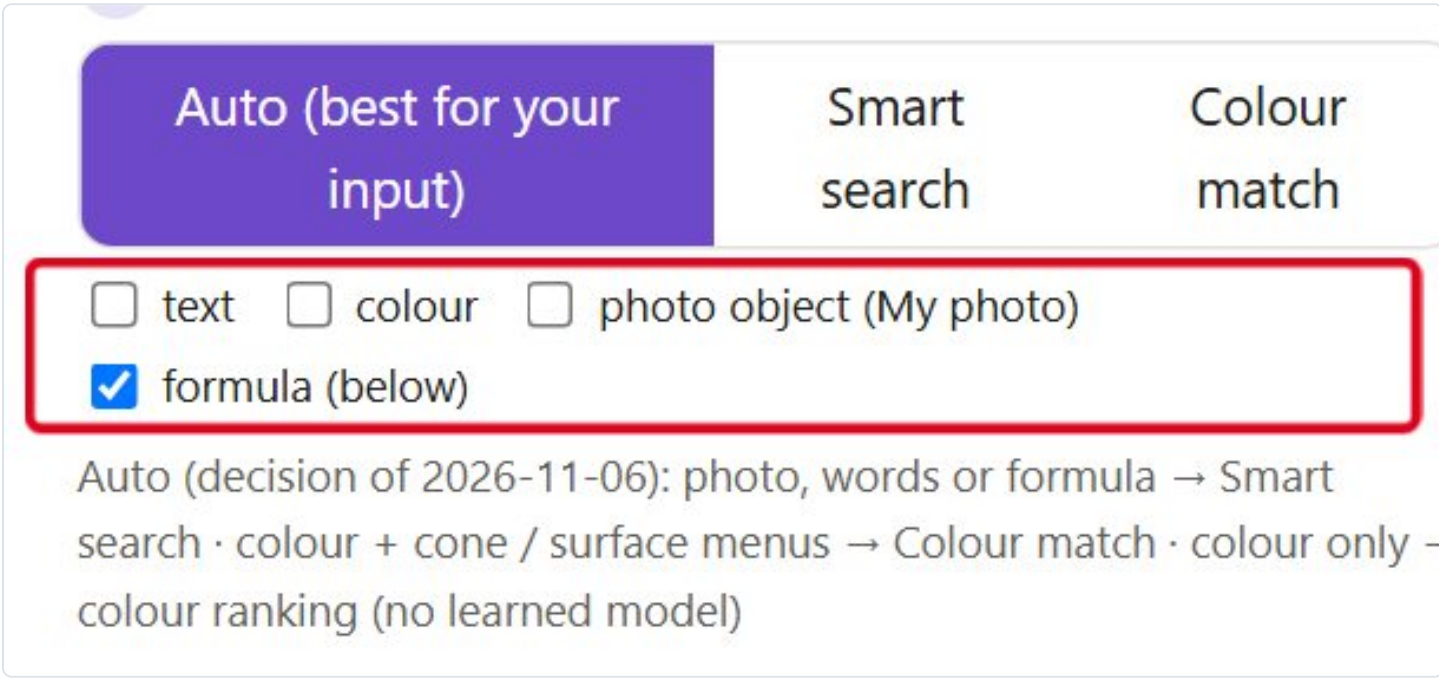
Read with care: this photo is in the database, so finding its own recipe (Satin matte base glaze, #614435) at #1 is expected, not proof. Its surface on Glazy is Satin-matte, not Satin, so the surface part is 0. The app took the large warm-grey tile (#8B7770); for the sand glaze, click the yellow tile and use 🎨 Pick colour.

Case 3c · Only a formula → Smart search on the formula

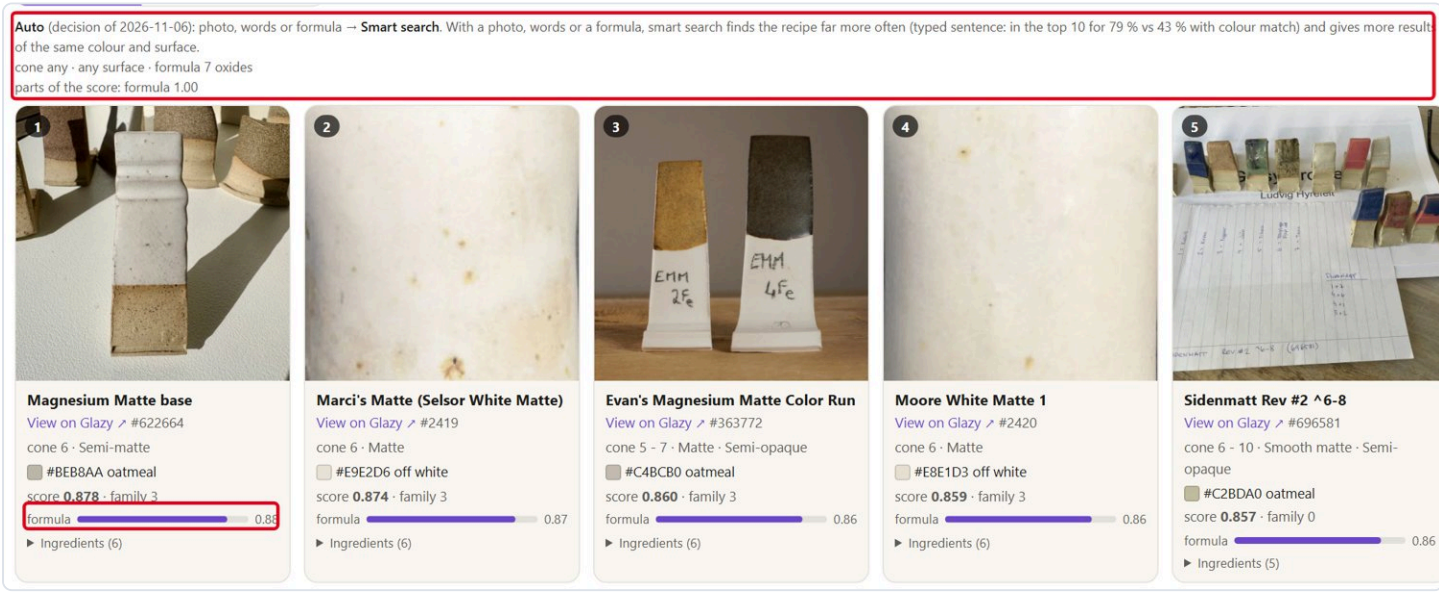
1. Untick text, colour and photo object in 4 · How to search.
2. Open the formula card, paste `SiO₂ 44.0 · Al₂O₃ 16.8 · CaO 9.7 · MgO 7.0 · Na₂O 2.9 · B₂O₃ 2.8 · K₂O 0.8` (the real formula of #614435, with subscripts) and press **Fill**: 'formula' is ticked automatically. Find recipes.



The formula card: the pasted analysis with subscripts, the 7 filled oxides and the check (framed).



Only 'formula' is ticked (framed).



Smart search on the formula alone: parts of the score = formula 1.00 (framed).

The calculation of #1

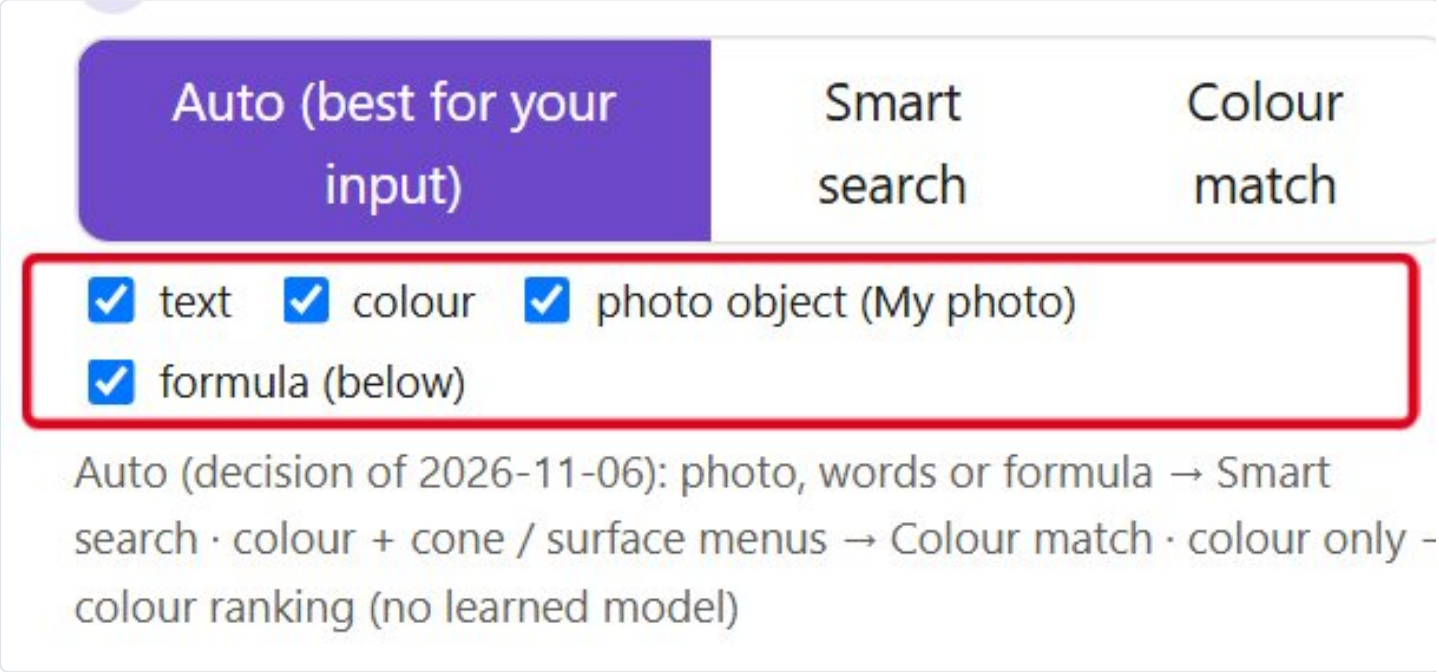
part	weight	value	weight × value	what the value is
formula	1.000	0.878	**0.878**	(cos + 1) / 2 of your formula and the recipe in the joint space
score of #1 Magnesium Matte base			**0.878**	

- **Magnesium Matte base** · 0.878 · oatmeal #BEB8AA · cone 6 · Semi-matte
- **Marci's Matte (Selsor White Matte)** · 0.874 · off white #E9E2D6 · cone 6 · Matte
- **Evan's Magnesium Matte Color Run** · 0.860 · oatmeal #C4BCB0 · cone 5 - 7 · Matte

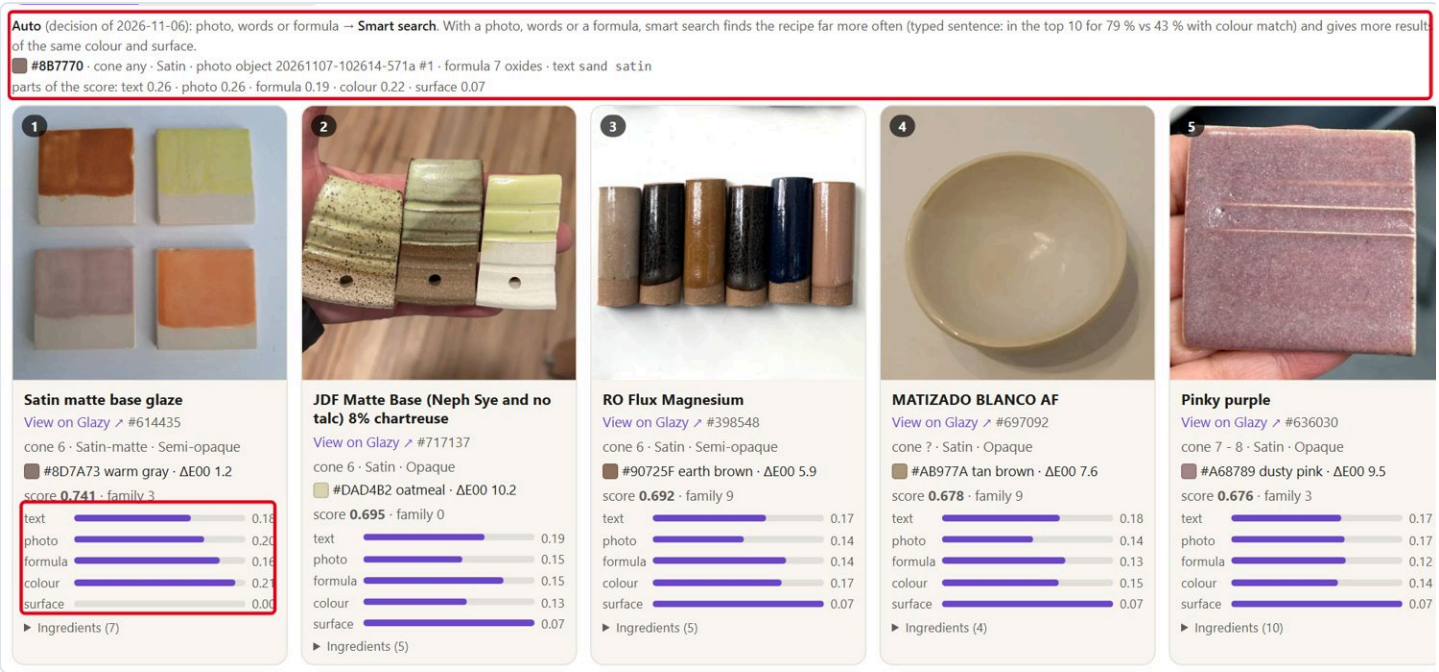
What it shows: the formula alone finds the **high-magnesium matte** family (MgO 7 %): every result is matte without any colour or surface given, because the chemistry predicts the surface. The recipe itself is not #1: the formula is compared with each recipe's combined text, photo and formula fingerprint.

Case 3d · Words + photo + formula → Smart search on everything

1. Tick text, colour, photo object and formula again; surface Satin. Find recipes.



All four ticked (framed).



Smart search with five parts (header framed); #1's bars (framed).

The calculation of #1

part	weight	value	weight × value	what the value is
words	0.259	0.676	**0.175**	(cos + 1) / 2 of your words and the recipe in the joint space
photo	0.259	0.762	**0.198**	(cos + 1) / 2 of the cut-out photo and the recipe in the joint space
formula	0.185	0.853	**0.158**	(cos + 1) / 2 of your formula and the recipe in the joint space
colour	0.222	0.944	**0.210**	exp(−ΔE00 / 20), ΔE00 = 1.16
surface	0.074	0.000	**0.000**	1 = matches, 0 = does not
score of #1 Satin matte base glaze			**0.741**	

- **Satin matte base glaze** · 0.741 · warm gray #8D7A73 · cone 6 · Satin-matte
- **JDF Matte Base (Neph Sye and no talc) 8% chartreuse** · 0.695 · oatmeal #DAD4B2 · cone 6 · Satin
- **RO Flux Magnesium** · 0.692 · earth brown #90725F · cone 6 · Satin

What it shows: with all three inputs the recipe itself is back at #1, now helped by its formula (0.853).

Summary

Case	You give	Auto uses	#1 (score)
1	only the colour sand	Colour ranking	Test 2 - Sub flux (Mag carb, no zinc, no calc carb) (0.770)
2a	colour + surface Satin	Colour match	Test 2 - Sub flux (borato calcio) (0.655)
2b	colour + Satin + cone 6	Colour match	MC6G HCSM2 - Beth's Test1 (Ru 7.2/ Cu 3.2/ Borate 10) (0.753)
3a	typed: sand satin	Smart search	^6 Charles Mingus Base V4 (with Neph Sy Sub for Cornwall & Lowered Si & Al) (0.817)
3b	+ a photo object	Smart search	Satin matte base glaze (0.715)
3c	only the formula	Smart search	Magnesium Matte base (0.878)
3d	words + photo + formula	Smart search	Satin matte base glaze (0.741)

- **Only a colour:** exact colour, nothing else controlled. **With the menus:** always set the cone.
- **Typed words** give the best list without a photo; with a photo, leave colour words out of the text (they replace the photo's colour).
- **A formula** needs at least 3 oxides including a flux; subscripts (SiO_2) are read. Alone, it finds the chemistry family.
- A database photo finds its own recipe: that is expected, not a test of the model.